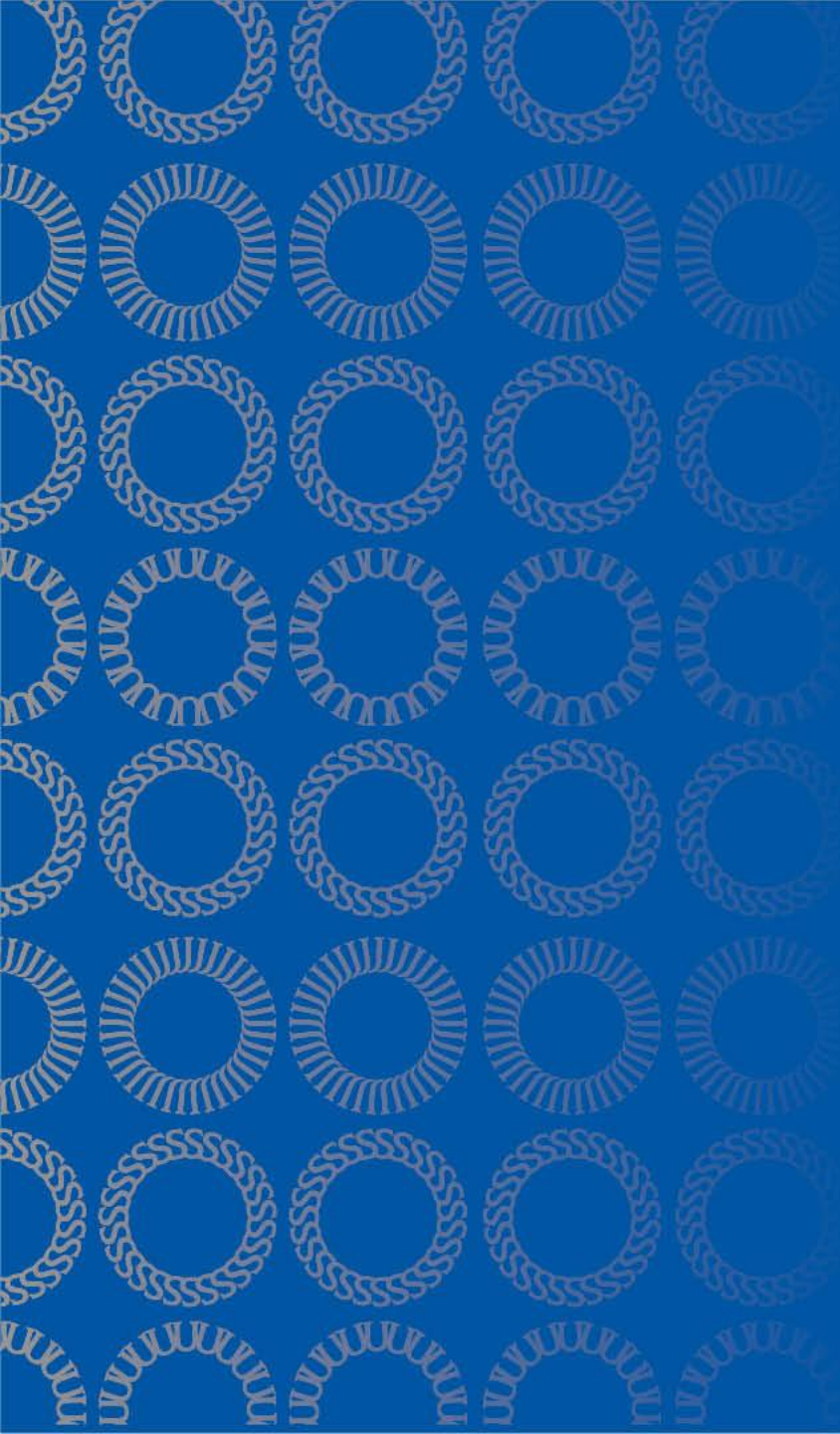




DATA 255

Deep Learning and Computer Vision in Applied Data Intelligence

Sept 16, 2026

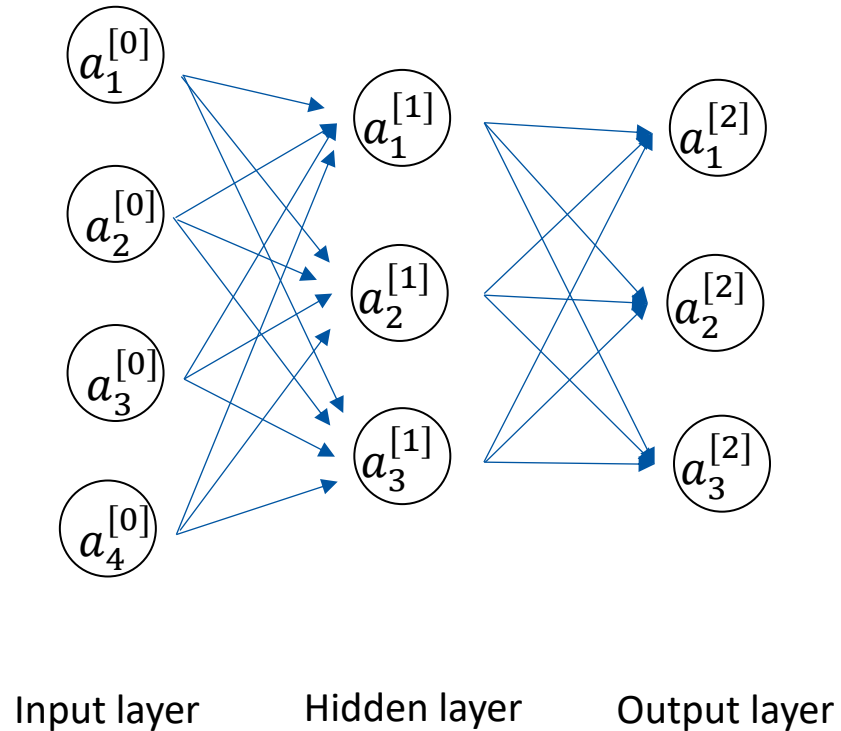
Taehee Jeong, Ph.D.

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Today's Topics

- Backward Pass for classification with Softmax function

Neural Network: 3 output neurons



Activation function of the last layer is *Softmax*.

Forward propagation

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

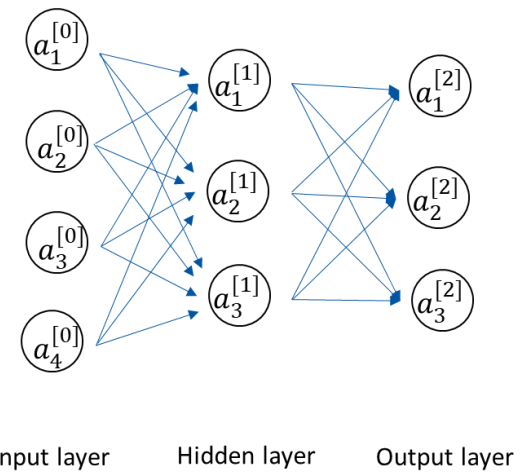
$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$



Forward propagation

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

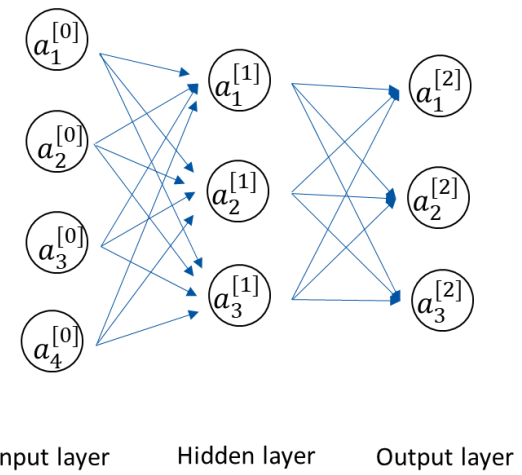
$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

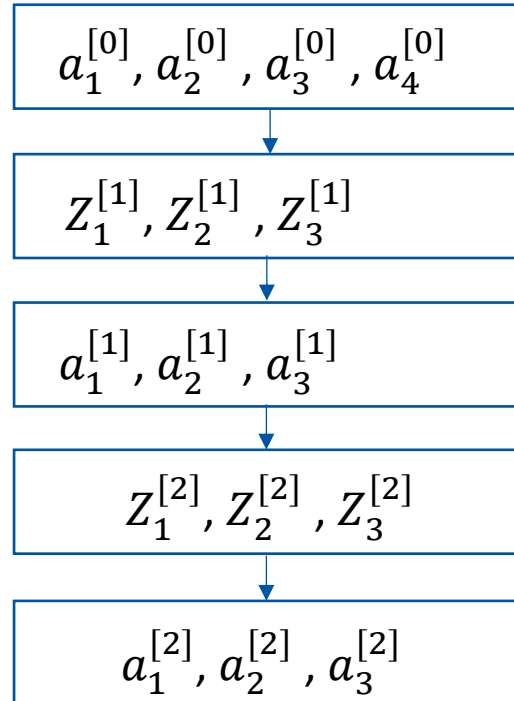
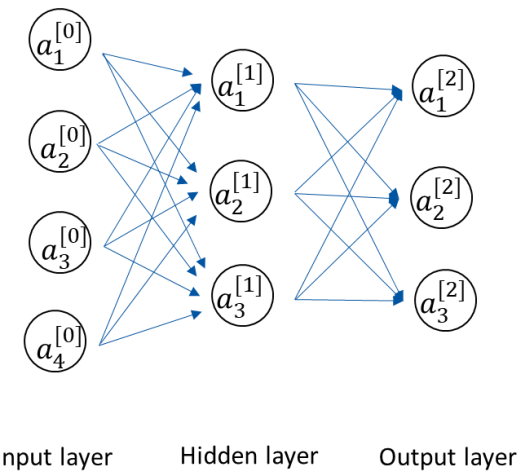
$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$



Forward propagation



Derivative of $a^{[2]}$

backpropagation for cross entropy loss

$$L(\hat{y}, y) = -[y_1 * \log \hat{y}_1 + y_2 * \log \hat{y}_2 + y_3 * \log \hat{y}_3]$$

$$L(\hat{y}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

Activation function of output layer is softmax

$$a_1^{[2]} = \frac{\exp(Z_1^{[2]})}{\sum_{j=1}^K \exp(Z_j^{[2]})}$$

$$a_1^{[2]} = \frac{\exp(Z_1^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$a_2^{[2]} = \frac{\exp(Z_2^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$a_3^{[2]} = \frac{\exp(Z_3^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

Derivative of *softmax*

$$S(Z_1) = \frac{\exp(Z_1)}{\exp(Z_1) + \exp(Z_2) + \exp(Z_3)}$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = S(Z_i) \times (1 - S(Z_i)) \quad \text{if } i = j$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = -S(Z_i) \times S(Z_j) \quad \text{if } i \neq j$$

Chain rule

$$h(x) = f(x) \cdot g(x)$$

$$\frac{d}{dx} h(x) = \left[\frac{d}{dx} f(x) \right] \cdot g(x) + f(x) \cdot \left[\frac{d}{dx} g(x) \right]$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{-x} = -e^{-x}$$

Derivative of *softmax*

$$S(Z_1) = \frac{\exp(Z_1)}{\exp(Z_1) + \exp(Z_2) + \exp(Z_3)}$$

$$\Sigma = \exp(Z_1) + \exp(Z_2) + \exp(Z_3)$$

$$\frac{\partial S(Z_1)}{\partial Z_1} = \frac{\exp(Z_1)}{\Sigma} - \exp(Z_1) \Sigma^{-2} \exp(Z_1)$$

$$= \frac{\exp(Z_1)}{\Sigma} - \frac{\exp(Z_1) \exp(Z_1)}{\Sigma^2}$$

$$= \frac{\exp(Z_1) \Sigma - \exp(Z_1) \exp(Z_1)}{\Sigma^2}$$

$$= \frac{\exp(Z_1)}{\Sigma} \times \frac{\Sigma - \exp(Z_1)}{\Sigma}$$

$$= S(Z_1) \times (1 - S(Z_1))$$

Chain rule

$$h(x) = f(x) \cdot g(x)$$

$$\frac{d}{dx} h(x) = \left[\frac{d}{dx} f(x) \right] \cdot g(x) + f(x) \cdot \left[\frac{d}{dx} g(x) \right]$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{-x} = -e^{-x}$$

Derivative of *softmax*

$$S(Z_2) = \frac{\exp(Z_2)}{\exp(Z_1) + \exp(Z_2) + \exp(Z_3)}$$

$$\Sigma = \exp(Z_1) + \exp(Z_2) + \exp(Z_3)$$

$$\frac{\partial S(Z_2)}{\partial Z_1} = \frac{0}{\Sigma} - \exp(Z_2) \Sigma^{-2} \exp(Z_1)$$

$$= - \frac{\exp(Z_2) \exp(Z_1)}{\Sigma^2}$$

$$= -S(Z_1) \times S(Z_2)$$

Chain rule

$$h(x) = f(x) \cdot g(x)$$

$$\frac{d}{dx} h(x) = \left[\frac{d}{dx} f(x) \right] \cdot g(x) + f(x) \cdot \left[\frac{d}{dx} g(x) \right]$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} e^{-x} = -e^{-x}$$

Derivative of $Z^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$L(a_i^{[2]}, y) = -[y_1 * \log(S(Z_1^{[2]})) + y_2 * \log(S(Z_2^{[2]})) + y_3 * \log(S(Z_3^{[2]}))]$$

$$dZ_1^{[2]} = \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}}$$

$$dZ_2^{[2]} = \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_1^{[2]}}$$

$$dZ_3^{[2]} = \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_1^{[2]}}$$

$$a_1^{[2]} = S(Z_1^{[2]}) = \frac{\exp(Z_1^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = S(Z_i) \times (1 - S(Z_i)) \quad \text{if } i = j$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = -S(Z_i) \times S(Z_j) \quad \text{if } i \neq j$$

Derivative of $dZ_1^{[2]}$

$$L(\hat{y}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$L(\hat{y}, y) = -[y_1 * \log(S(Z_1^{[2]})) + y_2 * \log(S(Z_2^{[2]})) + y_3 * \log(S(Z_3^{[2]}))]$$

$$dZ_1^{[2]} = -\frac{y_1}{a_1^{[2]}} \cdot \frac{\partial S(Z_1^{[2]})}{\partial Z_1^{[2]}} - \frac{y_2}{a_2^{[2]}} \cdot \frac{\partial S(Z_2^{[2]})}{\partial Z_1^{[2]}} - \frac{y_3}{a_3^{[2]}} \cdot \frac{\partial S(Z_3^{[2]})}{\partial Z_1^{[2]}}$$

$$dZ_1^{[2]} = -y_1 \cdot (1 - a_1^{[2]}) + y_2 \cdot a_1^{[2]} + y_3 \cdot a_1^{[2]}$$

$$dZ_1^{[2]} = -y_1 + y_1 \cdot a_1^{[2]} + y_2 \cdot a_1^{[2]} + y_3 \cdot a_1^{[2]}$$

$$dZ_1^{[2]} = -y_1 + a_1^{[2]}$$

$$dZ_1^{[2]} = a_1^{[2]} - y_1$$

$$a_1^{[2]} = S(Z_1^{[2]}) = \frac{\exp(Z_1^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = S(Z_i) \times (1 - S(Z_i))$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = -S(Z_i) \times S(Z_j)$$

Derivative of $dZ_2^{[2]}$

$$L(\hat{y}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$L(\hat{y}, y) = -[y_1 * \log(S(Z_1^{[2]})) + y_2 * \log(S(Z_2^{[2]})) + y_3 * \log(S(Z_3^{[2]}))]$$

$$dZ_2^{[2]} = -\frac{y_1}{a_1^{[2]}} \cdot \frac{\partial S(Z_1^{[2]})}{\partial Z_2^{[2]}} - \frac{y_2}{a_2^{[2]}} \cdot \frac{\partial S(Z_2^{[2]})}{\partial Z_2^{[2]}} - \frac{y_3}{a_3^{[2]}} \cdot \frac{\partial S(Z_3^{[2]})}{\partial Z_2^{[2]}}$$

$$dZ_2^{[2]} = y_1 \cdot a_2^{[2]} - y_2 \cdot (1 - a_2^{[2]}) + y_3 \cdot a_2^{[2]}$$

$$dZ_2^{[2]} = -y_2 + y_1 \cdot a_2^{[2]} + y_2 \cdot a_2^{[2]} + y_3 \cdot a_2^{[2]}$$

$$dZ_2^{[2]} = -y_2 + a_2^{[2]}$$

$$dZ_2^{[2]} = a_2^{[2]} - y_2$$

$$a_2^{[2]} = S(Z_2^{[2]}) = \frac{\exp(Z_2^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = S(Z_i) \times (1 - S(Z_i))$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = -S(Z_i) \times S(Z_j)$$

Derivative of $dZ_3^{[2]}$

$$L(\hat{y}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$L(\hat{y}, y) = -[y_1 * \log(S(Z_1^{[2]})) + y_2 * \log(S(Z_2^{[2]})) + y_3 * \log(S(Z_3^{[2]}))]$$

$$dZ_3^{[2]} = -\frac{y_1}{a_1^{[2]}} \cdot \frac{\partial S(Z_1^{[2]})}{\partial Z_3^{[2]}} - \frac{y_2}{a_2^{[2]}} \cdot \frac{\partial S(Z_2^{[2]})}{\partial Z_3^{[2]}} - \frac{y_3}{a_3^{[2]}} \cdot \frac{\partial S(Z_3^{[2]})}{\partial Z_3^{[2]}}$$

$$dZ_3^{[2]} = y_1 \cdot a_3^{[2]} + y_2 \cdot a_3^{[2]} - y_3 \cdot (1 - a_3^{[2]})$$

$$dZ_3^{[2]} = -y_3 + y_1 \cdot a_3^{[2]} + y_2 \cdot a_3^{[2]} + y_3 \cdot a_3^{[2]}$$

$$dZ_3^{[2]} = -y_3 + a_3^{[2]}$$

$$dZ_3^{[2]} = a_3^{[2]} - y_3$$

$$a_3^{[2]} = S(Z_3^{[2]}) = \frac{\exp(Z_3^{[2]})}{\exp(Z_1^{[2]}) + \exp(Z_2^{[2]}) + \exp(Z_3^{[2]})}$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = S(Z_i) \times (1 - S(Z_i))$$

$$\frac{\partial S(Z_j)}{\partial Z_i} = -S(Z_i) \times S(Z_j)$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[2]}} = dW_{10}^{[2]}$$

$$\begin{aligned} dW_{10}^{[2]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial W_{10}^{[2]}} \\ &= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial W_{10}^{[2]}} \end{aligned}$$

$$dW_{10}^{[2]} = dZ_1^{[2]} \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{11}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{11}^{[2]}} = dW_{11}^{[2]}$$

$$\begin{aligned} dW_{11}^{[2]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial W_{11}^{[2]}} \\ &= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial W_{11}^{[2]}} \end{aligned}$$

$$dW_{11}^{[2]} = dZ_1^{[2]} \cdot a_1^{[1]}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{12}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{12}^{[2]}} = dW_{12}^{[2]}$$

$$\begin{aligned} dW_{12}^{[2]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial W_{12}^{[2]}} \\ &= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial W_{12}^{[2]}} \end{aligned}$$

$$dW_{12}^{[2]} = dZ_1^{[2]} \cdot a_2^{[1]}$$

Derivative of $W_{13}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{13}^{[2]}} = dW_{13}^{[2]}$$

$$\begin{aligned} dW_{13}^{[2]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial W_{13}^{[2]}} \\ &= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial W_{13}^{[2]}} \end{aligned}$$

$$dW_{13}^{[2]} = dZ_1^{[2]} \cdot a_3^{[1]}$$

Derivative of $W_{20}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{20}^{[2]}} = dW_{20}^{[2]}$$

$$\begin{aligned} dW_{20}^{[2]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial W_{20}^{[2]}} \\ &= dZ_2^{[2]} \frac{\partial Z_2^{[2]}}{\partial W_{20}^{[2]}} \end{aligned}$$

$$dW_{20}^{[2]} = dZ_2^{[2]} \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{21}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{21}^{[2]}} = dW_{21}^{[2]}$$

$$\begin{aligned} dW_{21}^{[2]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial W_{21}^{[2]}} \\ &= dZ_2^{[2]} \frac{\partial Z_2^{[2]}}{\partial W_{21}^{[2]}} \end{aligned}$$

$$dW_{21}^{[2]} = dZ_2^{[2]} \cdot a_1^{[1]}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $W_{22}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{22}^{[2]}} = dW_{22}^{[2]}$$

$$\begin{aligned} dW_{22}^{[2]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial W_{22}^{[2]}} \\ &= dZ_2^{[2]} \frac{\partial Z_2^{[2]}}{\partial W_{22}^{[2]}} \end{aligned}$$

$$dW_{22}^{[2]} = dZ_2^{[2]} \cdot a_2^{[1]}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $W_{23}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{23}^{[2]}} = dW_{23}^{[2]}$$

$$\begin{aligned} dW_{23}^{[2]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial W_{23}^{[2]}} \\ &= dZ_2^{[2]} \frac{\partial Z_2^{[2]}}{\partial W_{23}^{[2]}} \end{aligned}$$

$$dW_{23}^{[2]} = dZ_2^{[2]} \cdot a_3^{[1]}$$

Derivative of $W_{30}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{30}^{[2]}} = dW_{30}^{[2]}$$

$$\begin{aligned} dW_{30}^{[2]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial W_{30}^{[2]}} \\ &= dZ_3^{[2]} \frac{\partial Z_3^{[2]}}{\partial W_{30}^{[2]}} \end{aligned}$$

$$dW_{30}^{[2]} = dZ_3^{[2]} \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{31}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{31}^{[2]}} = dW_{31}^{[2]}$$

$$\begin{aligned} dW_{31}^{[2]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial W_{31}^{[2]}} \\ &= dZ_3^{[2]} \frac{\partial Z_3^{[2]}}{\partial W_{31}^{[2]}} \end{aligned}$$

$$dW_{31}^{[2]} = dZ_3^{[2]} \cdot a_1^{[1]}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $W_{32}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{32}^{[2]}} = dW_{32}^{[2]}$$

$$\begin{aligned} dW_{32}^{[2]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial W_{32}^{[2]}} \\ &= dZ_3^{[2]} \frac{\partial Z_3^{[2]}}{\partial W_{32}^{[2]}} \end{aligned}$$

$$dW_{32}^{[2]} = dZ_3^{[2]} \cdot a_2^{[1]}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $W_{33}^{[2]}$

$$L(a_i^{[2]}, y) = -[y_1 * \log(a_1^{[2]}) + y_2 * \log(a_2^{[2]}) + y_3 * \log(a_3^{[2]})]$$

$$\frac{\partial L(a, y)}{\partial W_{33}^{[2]}} = dW_{33}^{[2]}$$

$$\begin{aligned} dW_{33}^{[2]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial W_{33}^{[2]}} \\ &= dZ_3^{[2]} \frac{\partial Z_3^{[2]}}{\partial W_{33}^{[2]}} \end{aligned}$$

$$dW_{33}^{[2]} = dZ_3^{[2]} \cdot a_3^{[1]}$$

Derivative of $da_1^{[1]}$

$$\frac{\partial L(a, y)}{\partial a_1^{[1]}} = da_1^{[1]}$$

$$da_1^{[1]} = \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}}$$

$$= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}}$$

$$da_1^{[1]} = dZ_1^{[2]} \cdot W_{11}^{[2]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $da_1^{[1]}$

$$\frac{\partial L(a, y)}{\partial a_1^{[1]}} = da_1^{[1]}$$

$$da_1^{[1]} = \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial a_1^{[1]}}$$

$$= dZ_2^{[2]} \frac{\partial Z_2^{[2]}}{\partial a_1^{[1]}}$$

$$da_1^{[1]} = dZ_2^{[2]} \cdot W_{21}^{[2]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $da_1^{[1]}$

$$\frac{\partial L(a, y)}{\partial a_1^{[1]}} = da_1^{[1]}$$

$$\begin{aligned} da_1^{[1]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial a_1^{[1]}} \\ &= dZ_3^{[2]} \frac{\partial Z_3^{[2]}}{\partial a_1^{[1]}} \end{aligned}$$

$$da_1^{[1]} = dZ_3^{[2]} \cdot W_{31}^{[2]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $da_2^{[1]}$

$$\frac{\partial L(a, y)}{\partial a_2^{[1]}} = da_2^{[1]}$$

$$da_2^{[1]} = \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}}$$

$$= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}}$$

$$da_1^{[1]} = dZ_1^{[2]} \cdot W_{12}^{[2]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $da_3^{[1]}$

$$\frac{\partial L(a, y)}{\partial a_3^{[1]}} = da_3^{[1]}$$

$$da_3^{[1]} = \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}}$$

$$= dZ_1^{[2]} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}}$$

$$da_1^{[1]} = dZ_1^{[2]} \cdot W_{13}^{[2]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = S(Z_1^{[2]})$$

$$a_2^{[2]} = S(Z_2^{[2]})$$

$$a_3^{[2]} = S(Z_3^{[2]})$$

Derivative of $dZ_1^{[1]}$

$$\frac{\partial L(a, y)}{\partial Z_1^{[1]}} = dZ_1^{[1]}$$

$$dZ_1^{[1]} = \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}$$

$$= da_1^{[1]} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}}$$

$$dZ_1^{[1]} = da_1^{[1]} * a_1^{[1]} * [1 - a_1^{[1]}]$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $dZ_2^{[1]}$

$$\begin{aligned}\frac{\partial L(\hat{y}, y)}{\partial Z_2^{[1]}} &= dZ_2^{[1]} \\ dZ_2^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \\ &= da_2^{[1]} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}}\end{aligned}$$

$$dZ_1^{[1]} = da_2^{[1]} * a_2^{[1]} * [1 - a_2^{[1]}]$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $dZ_3^{[1]}$

$$\begin{aligned}\frac{\partial L(a, y)}{\partial Z_3^{[1]}} &= dZ_3^{[1]} \\ dZ_3^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \\ &= da_3^{[1]} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}}\end{aligned}$$

$$dZ_3^{[1]} = da_3^{[1]} * a_3^{[1]} * [1 - a_3^{[1]}]$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \end{aligned}$$

$$dW_{10}^{[1]} = dZ_1^{[1]} \cdot 1$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \end{aligned}$$

$$dW_{10}^{[1]} = dZ_1^{[1]} \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \end{aligned}$$

$$dW_{10}^{[1]} = dZ_1^{[1]} \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= [a_1^{[2]} - y_1] \cdot W_{11}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1 \end{aligned}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_2^{[2]}} \frac{\partial a_2^{[2]}}{\partial Z_2^{[2]}} \frac{\partial Z_2^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= [a_2^{[2]} - y_2] \cdot W_{21}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1 \end{aligned}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{10}^{[1]}} = dW_{10}^{[1]}$$

$$\begin{aligned} dW_{10}^{[1]} &= \frac{\partial L}{\partial a_3^{[2]}} \frac{\partial a_3^{[2]}}{\partial Z_3^{[2]}} \frac{\partial Z_3^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{10}^{[1]}} \\ &= [a_3^{[2]} - y_3] \cdot W_{31}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1 \end{aligned}$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{10}^{[1]}$

$$dW_{10}^{[1]} = [a_1^{[2]} - y_1] \cdot W_{11}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1$$

$$dW_{10}^{[1]} = [a_2^{[2]} - y_2] \cdot W_{21}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1$$

$$dW_{10}^{[1]} = [a_3^{[2]} - y_3] \cdot W_{31}^{[2]} * a_1^{[1]} * [1 - a_1^{[1]}] \cdot 1$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{11}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{11}^{[1]}} = dW_{11}^{[1]}$$

$$\begin{aligned} dW_{11}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{11}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{11}^{[1]}} \end{aligned}$$

$$dW_{11}^{[1]} = dZ_1^{[1]} \cdot a_1^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{12}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{12}^{[1]}} = dW_{12}^{[1]}$$

$$\begin{aligned} dW_{12}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{12}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{12}^{[1]}} \end{aligned}$$

$$dW_{12}^{[1]} = dZ_1^{[1]} \cdot a_2^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{13}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{13}^{[1]}} = dW_{13}^{[1]}$$

$$\begin{aligned} dW_{13}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{13}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{13}^{[1]}} \end{aligned}$$

$$dW_{13}^{[1]} = dZ_1^{[1]} \cdot a_3^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{14}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{14}^{[1]}} = dW_{14}^{[1]}$$

$$\begin{aligned} dW_{14}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_1^{[1]}} \frac{\partial a_1^{[1]}}{\partial Z_1^{[1]}} \frac{\partial Z_1^{[1]}}{\partial W_{14}^{[1]}} \\ &= dZ_1^{[1]} \frac{\partial Z_1^{[1]}}{\partial W_{14}^{[1]}} \end{aligned}$$

$$dW_{14}^{[1]} = dZ_1^{[1]} \cdot a_4^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{20}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{20}^{[1]}} = dW_{20}^{[1]}$$

$$\begin{aligned} dW_{20}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \frac{\partial Z_2^{[1]}}{\partial W_{20}^{[1]}} \\ &= dZ_2^{[1]} \frac{\partial Z_2^{[1]}}{\partial W_{20}^{[1]}} \end{aligned}$$

$$dW_{20}^{[1]} = dZ_2^{[1]} \cdot 1$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{21}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{21}^{[1]}} = dW_{21}^{[1]}$$

$$\begin{aligned} dW_{21}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \frac{\partial Z_2^{[1]}}{\partial W_{21}^{[1]}} \\ &= dZ_2^{[1]} \frac{\partial Z_2^{[1]}}{\partial W_{21}^{[1]}} \end{aligned}$$

$$dW_{21}^{[1]} = dZ_2^{[1]} \cdot a_1^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{22}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{22}^{[1]}} = dW_{22}^{[1]}$$

$$\begin{aligned} dW_{22}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \frac{\partial Z_2^{[1]}}{\partial W_{22}^{[1]}} \\ &= dZ_2^{[1]} \frac{\partial Z_2^{[1]}}{\partial W_{22}^{[1]}} \end{aligned}$$

$$dW_{22}^{[1]} = dZ_2^{[1]} \cdot a_2^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{23}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{23}^{[1]}} = dW_{23}^{[1]}$$

$$\begin{aligned} dW_{23}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \frac{\partial Z_2^{[1]}}{\partial W_{23}^{[1]}} \\ &= dZ_2^{[1]} \frac{\partial Z_2^{[1]}}{\partial W_{23}^{[1]}} \end{aligned}$$

$$dW_{23}^{[1]} = dZ_2^{[1]} \cdot a_3^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{24}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{24}^{[1]}} = dW_{24}^{[1]}$$

$$\begin{aligned} dW_{24}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_2^{[1]}} \frac{\partial a_2^{[1]}}{\partial Z_2^{[1]}} \frac{\partial Z_2^{[1]}}{\partial W_{24}^{[1]}} \\ &= dZ_2^{[1]} \frac{\partial Z_2^{[1]}}{\partial W_{24}^{[1]}} \end{aligned}$$

$$dW_{24}^{[1]} = dZ_2^{[1]} \cdot a_4^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{30}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{30}^{[1]}} = dW_{30}^{[1]}$$

$$\begin{aligned} dW_{30}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \frac{\partial Z_3^{[1]}}{\partial W_{30}^{[1]}} \\ &= dZ_3^{[1]} \frac{\partial Z_3^{[1]}}{\partial W_{30}^{[1]}} \end{aligned}$$

$$dW_{30}^{[1]} = dZ_3^{[1]} \cdot 1$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

2nd layer

$$Z_1^{[2]} = W_{10}^{[2]} + W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + W_{13}^{[2]} a_3^{[1]}$$

$$Z_2^{[2]} = W_{20}^{[2]} + W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + W_{23}^{[2]} a_3^{[1]}$$

$$Z_3^{[2]} = W_{30}^{[2]} + W_{31}^{[2]} a_1^{[1]} + W_{32}^{[2]} a_2^{[1]} + W_{33}^{[2]} a_3^{[1]}$$

$$a_1^{[2]} = s(Z_1^{[2]})$$

$$a_2^{[2]} = s(Z_2^{[2]})$$

$$a_3^{[2]} = s(Z_3^{[2]})$$

Derivative of $W_{31}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{31}^{[1]}} = dW_{31}^{[1]}$$

$$\begin{aligned} dW_{31}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \frac{\partial Z_3^{[1]}}{\partial W_{31}^{[1]}} \\ &= dZ_3^{[1]} \frac{\partial Z_3^{[1]}}{\partial W_{31}^{[1]}} \end{aligned}$$

$$dW_{31}^{[1]} = dZ_3^{[1]} \cdot a_1^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{32}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{32}^{[1]}} = dW_{32}^{[1]}$$

$$\begin{aligned} dW_{32}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \frac{\partial Z_3^{[1]}}{\partial W_{32}^{[1]}} \\ &= dZ_3^{[1]} \frac{\partial Z_3^{[1]}}{\partial W_{32}^{[2]}} \end{aligned}$$

$$dW_{32}^{[2]} = dZ_3^{[1]} \cdot a_2^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{33}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{33}^{[1]}} = dW_{33}^{[1]}$$

$$\begin{aligned} dW_{33}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \frac{\partial Z_3^{[1]}}{\partial W_{33}^{[1]}} \\ &= dZ_3^{[1]} \frac{\partial Z_3^{[1]}}{\partial W_{33}^{[1]}} \end{aligned}$$

$$dW_{33}^{[1]} = dZ_3^{[1]} \cdot a_3^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Derivative of $W_{34}^{[1]}$

$$\frac{\partial L(a, y)}{\partial W_{34}^{[1]}} = dW_{34}^{[1]}$$

$$\begin{aligned} dW_{34}^{[1]} &= \frac{\partial L}{\partial a_1^{[2]}} \frac{\partial a_1^{[2]}}{\partial Z_1^{[2]}} \frac{\partial Z_1^{[2]}}{\partial a_3^{[1]}} \frac{\partial a_3^{[1]}}{\partial Z_3^{[1]}} \frac{\partial Z_3^{[1]}}{\partial W_{34}^{[1]}} \\ &= dZ_3^{[1]} \frac{\partial Z_3^{[1]}}{\partial W_{34}^{[1]}} \end{aligned}$$

$$dW_{34}^{[1]} = dZ_3^{[1]} \cdot a_4^{[0]}$$

1st layer

$$Z_1^{[1]} = W_{10}^{[1]} + W_{11}^{[1]} a_1^{[0]} + W_{12}^{[1]} a_2^{[0]} + W_{13}^{[1]} a_3^{[0]} + W_{14}^{[1]} a_4^{[0]}$$

$$Z_2^{[1]} = W_{20}^{[1]} + W_{21}^{[1]} a_1^{[0]} + W_{22}^{[1]} a_2^{[0]} + W_{23}^{[1]} a_3^{[0]} + W_{24}^{[1]} a_4^{[0]}$$

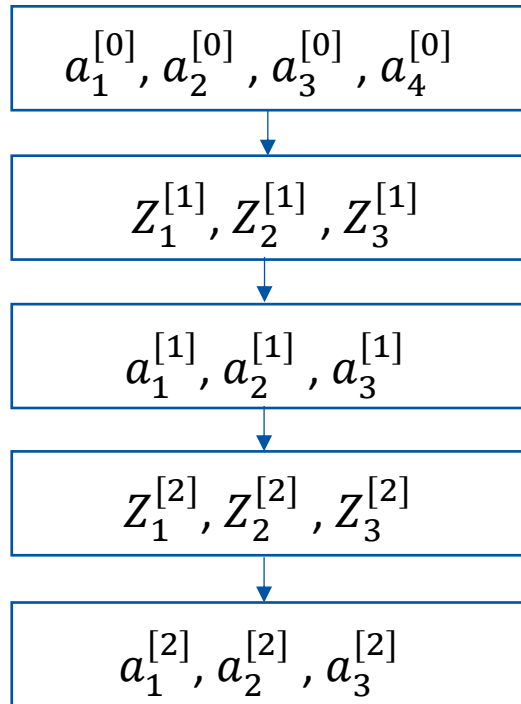
$$Z_3^{[1]} = W_{30}^{[1]} + W_{31}^{[1]} a_1^{[0]} + W_{32}^{[1]} a_2^{[0]} + W_{33}^{[1]} a_3^{[0]} + W_{34}^{[1]} a_4^{[0]}$$

$$a_1^{[1]} = \sigma(Z_1^{[1]})$$

$$a_2^{[1]} = \sigma(Z_2^{[1]})$$

$$a_3^{[1]} = \sigma(Z_3^{[1]})$$

Forward propagation



Backward propagation

